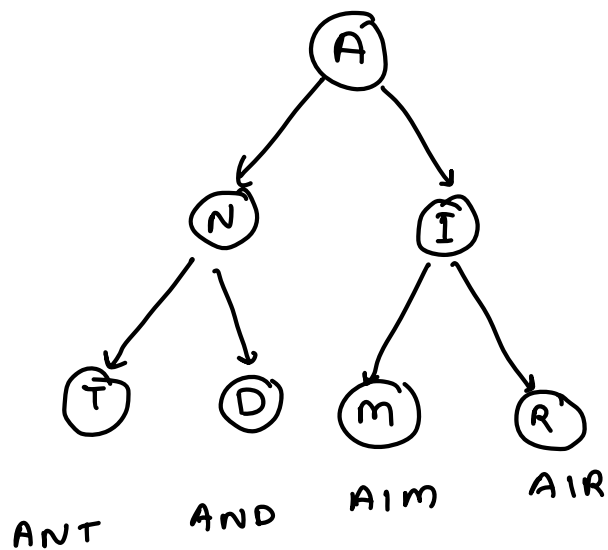
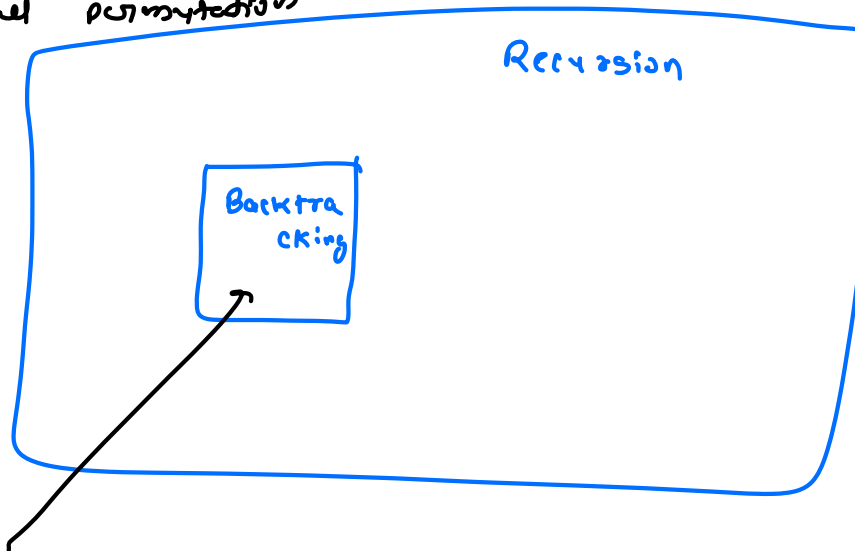


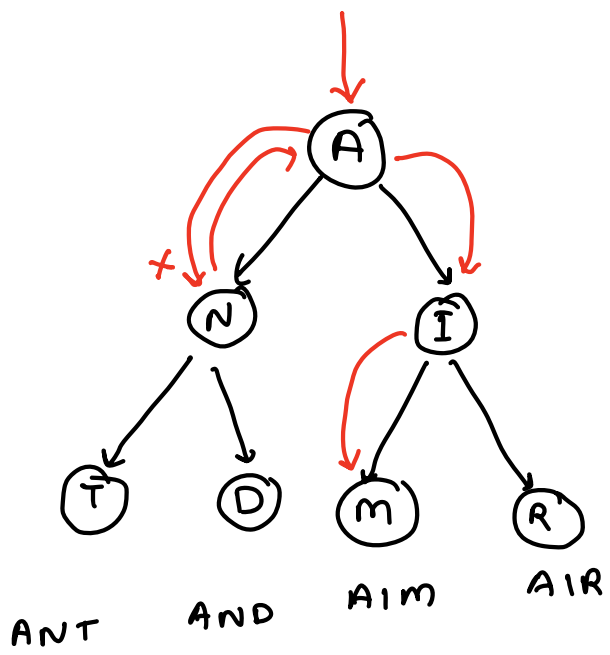
Agenda

Backtracking

[Recursion]

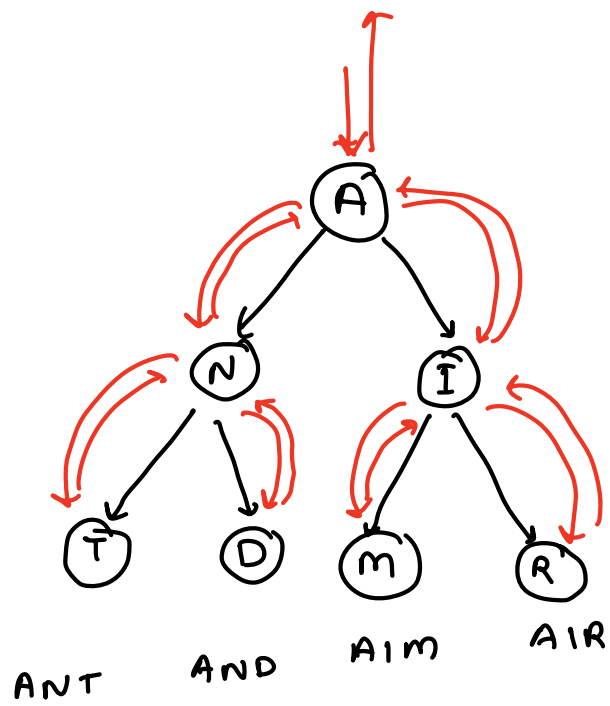
- ✓ 1. Backtracking
- ✓ 2. print all valid brackets
- ✓ 3. print all subsets
- ✓ 4. print all permutations





↓
AIM

ans = true



ANT
AND
AIM
AIR

Q. Given integer A. print all valid parentheses sequences containing A pair of parentheses

$$A \approx 2$$

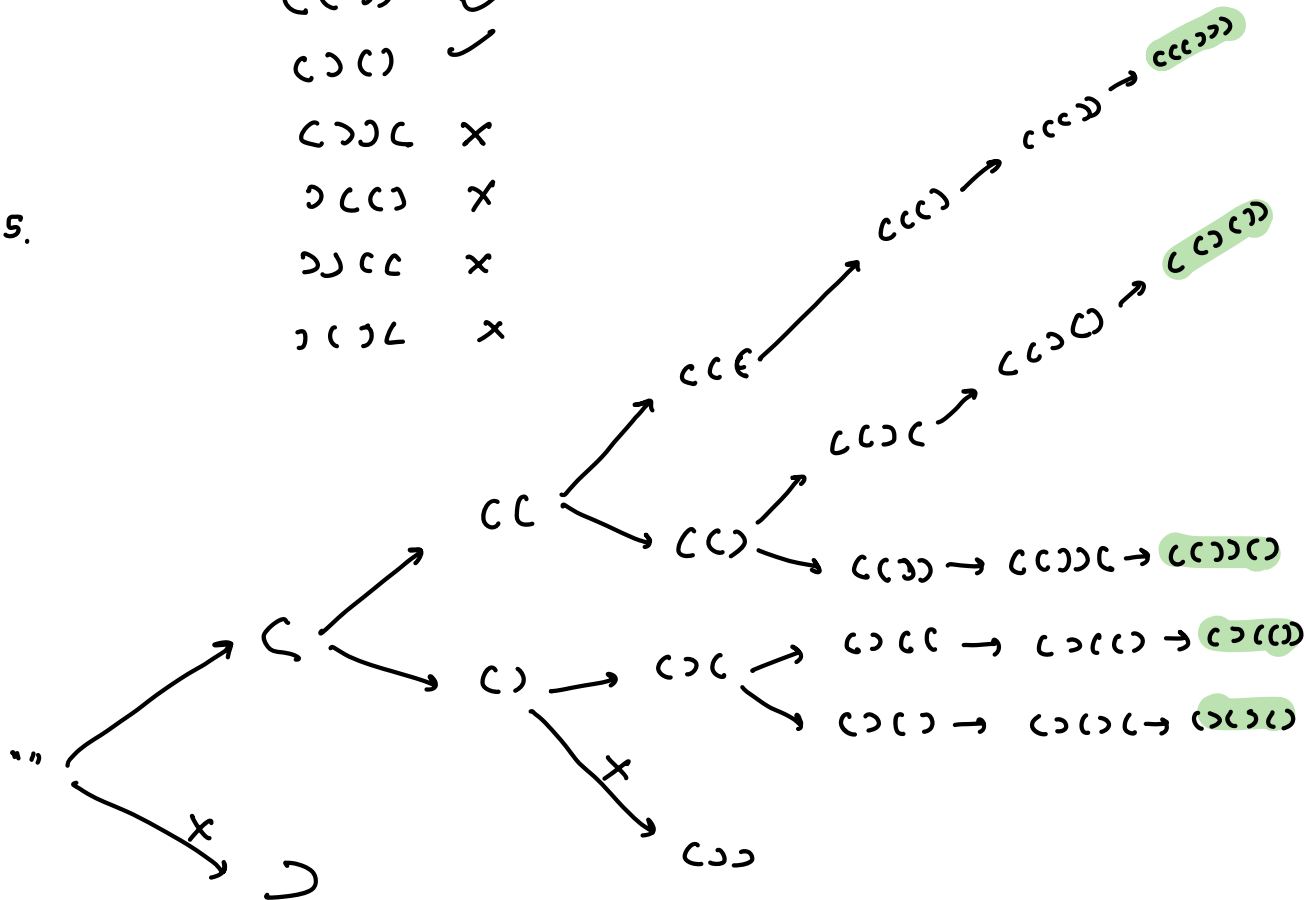
() ()

Ans = 5.

A = 3

(())	✓
() ()	✓
() () ()	x
() () ()	x
() () () ()	x
() () () ()	x

ans = 2



void Solve (string curr, int 0, int c, int A)

no of
pairs

if (c == A)

{
 print (curr)
 return
}

if (0 < A)

{
 solve (curr + 'c', 0 + 1, c, A)
}

if (0 > c)

{
 solve (curr + 's', 0, c + 1, A)
}

Solve ("", 0, 0)

A = 2

if (0 < A)

{ Solve ("c", 1, 0)

if (0 > 0) X

Solve ("c", 1, 0)

if (0 < A)
1 < 2

{ Solve ("cc", 2, 0)

if (1 > 0)

{ Solve ("c", 1, 1)

Solve ("cc", 2, 0)

if (2 < 2) X

{ X

if (2 > 0)

{ Solve ("ccc", 3, 1)

Solve ("c", 1, 1)

if (1 < A)

{ Solve ("cc", 2, 1)

if (1 > 1) X

Solve ("cc", 2, 1)

if (2 < 2) X

if (2 > 1)

{ Solve ("ccc", 3, 2)

Solve ()

Base (2 == 2)

Solve ("cc", 2, 1)

if (2 < A) X

if (2 > 1)

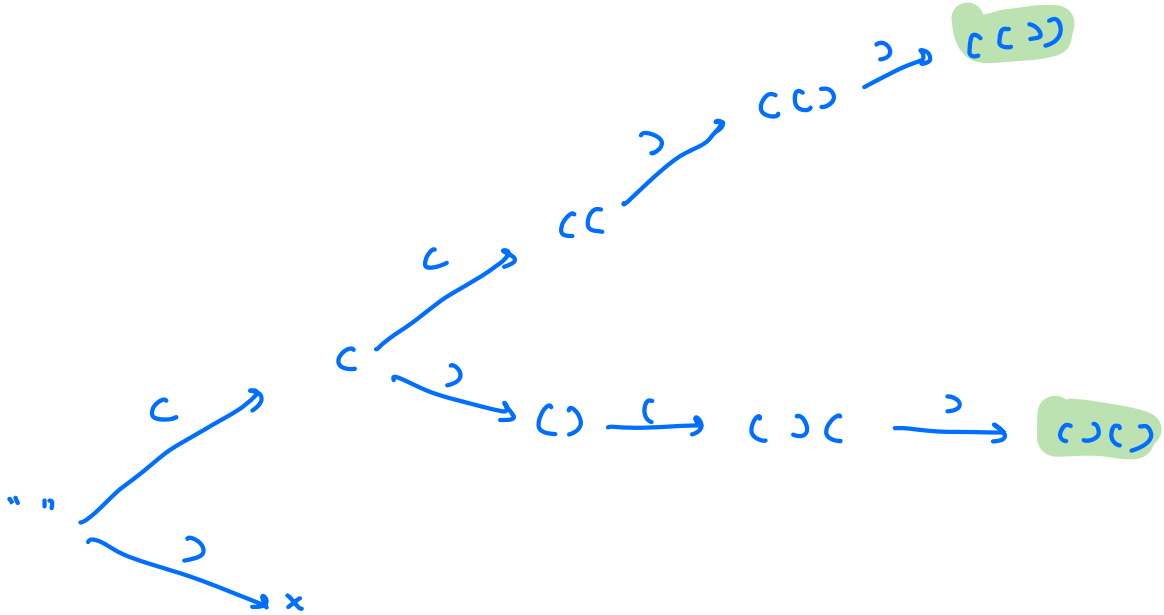
{ Solve ("ccc", 3, 2)

$$\{$$

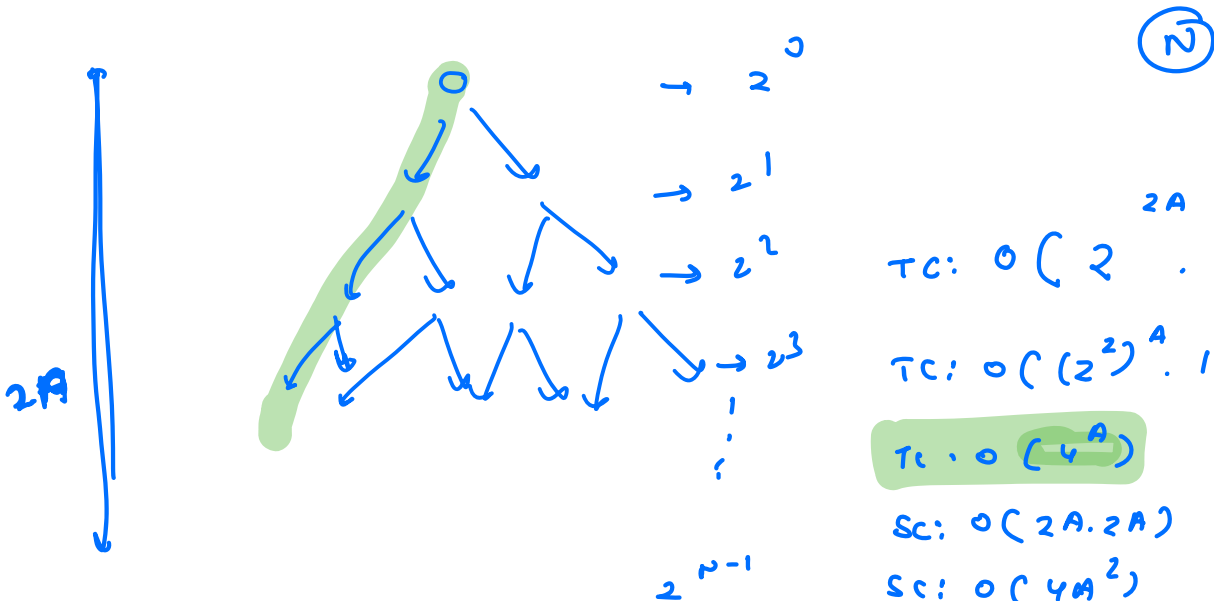
∫ பதிக
உயம்

140 2 > 20 x

ככ ככ
ככ ככ



TC. :


$$TC: 0 \begin{pmatrix} 2 & 1 \end{pmatrix}^{2A}$$
$$\tau_C: \mathcal{O}(z^2)^4 \rightarrow 1)$$
$$T_c = 0 \text{ (4A)}$$

Sc: $\circ (2A, 2A)$

$$Sc: 0(4A^2)$$
$$Sc: \partial(A^2)$$

void f ()

Base condition

for each option

// do something [empty]

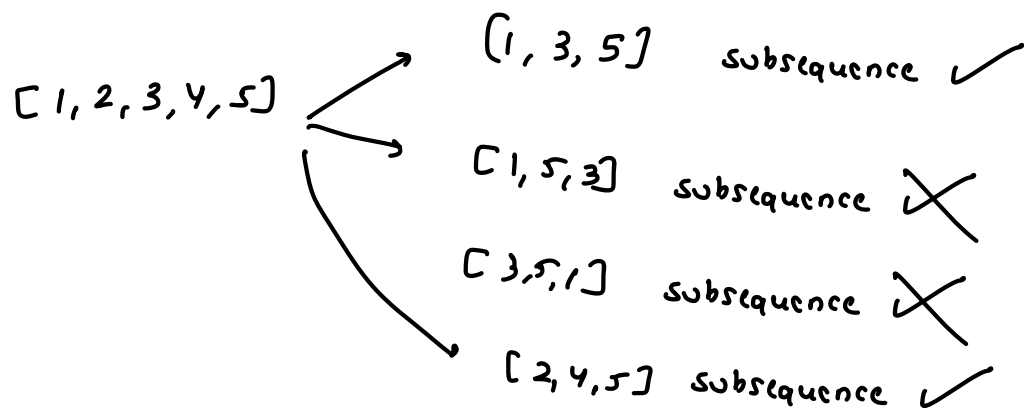
f (recursive)

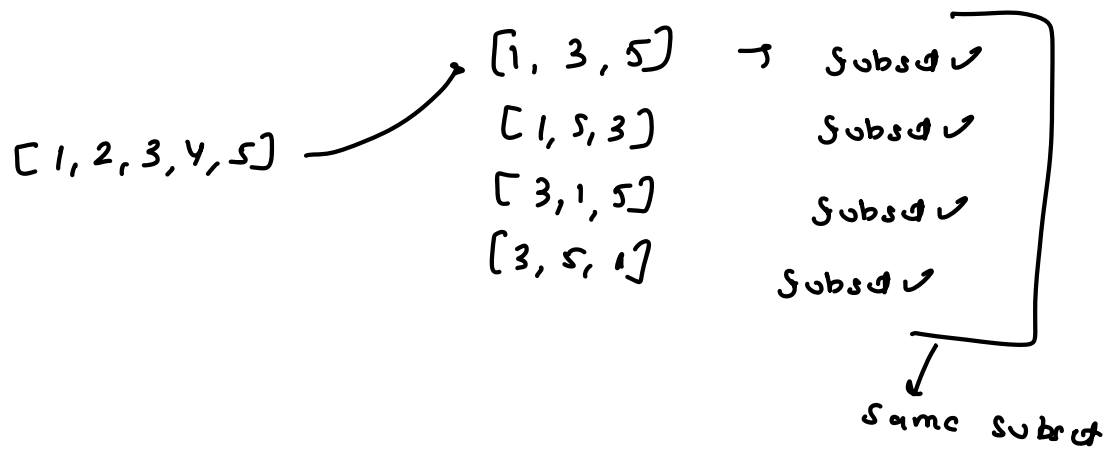
// undo

Subset & Subsequence

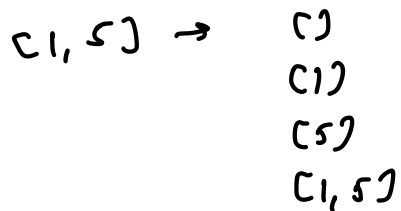
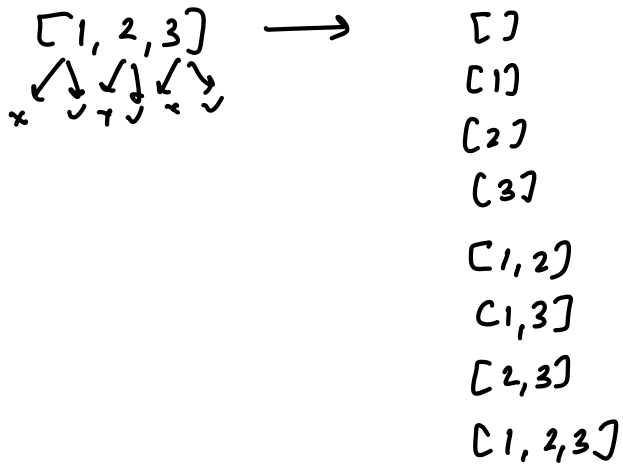
[order doesn't
matter]

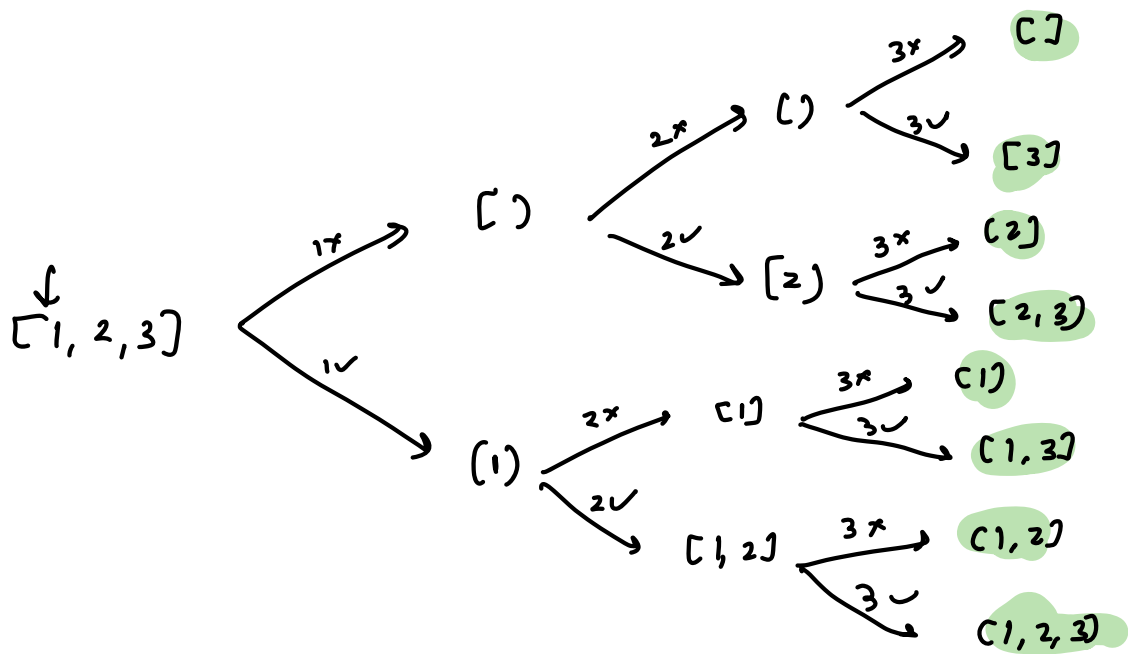
[order matters, needs to be
same order as input]





Q. Given array print all subseqs.





```
void solve ( A[], int curr, ArrayList currsubset)
```

```
    if ( curr == N)
```

```
    {
        print (currsubset)
        return
    }
```

```
// option 1 take currth
```

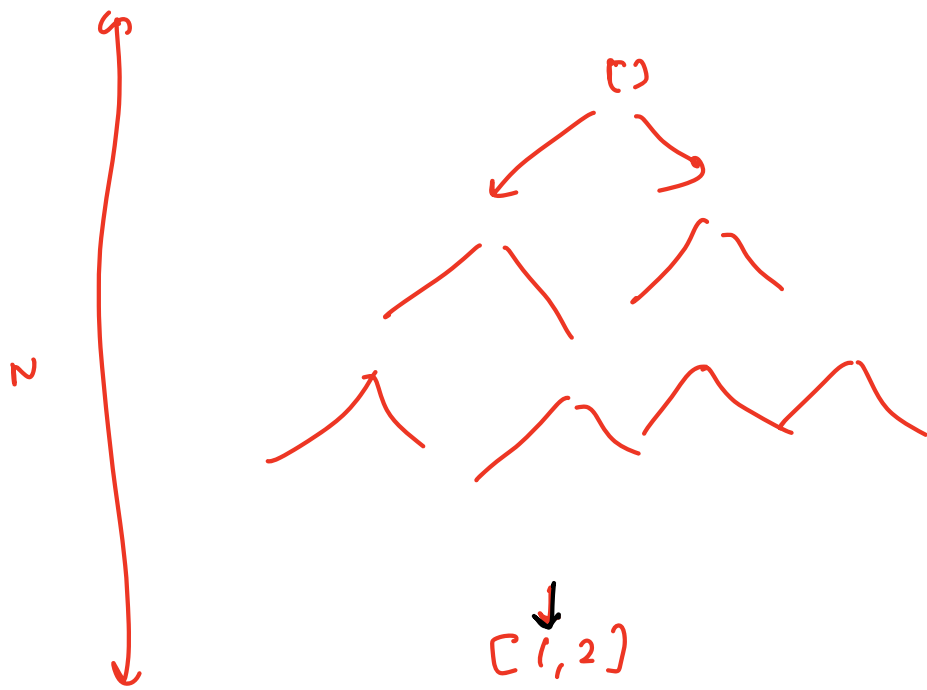
```
    currsubset.add ( A[curr])
```

```
    solve ( A, curr+1, currsubset)
```

```
    currsubset.remove ()
```

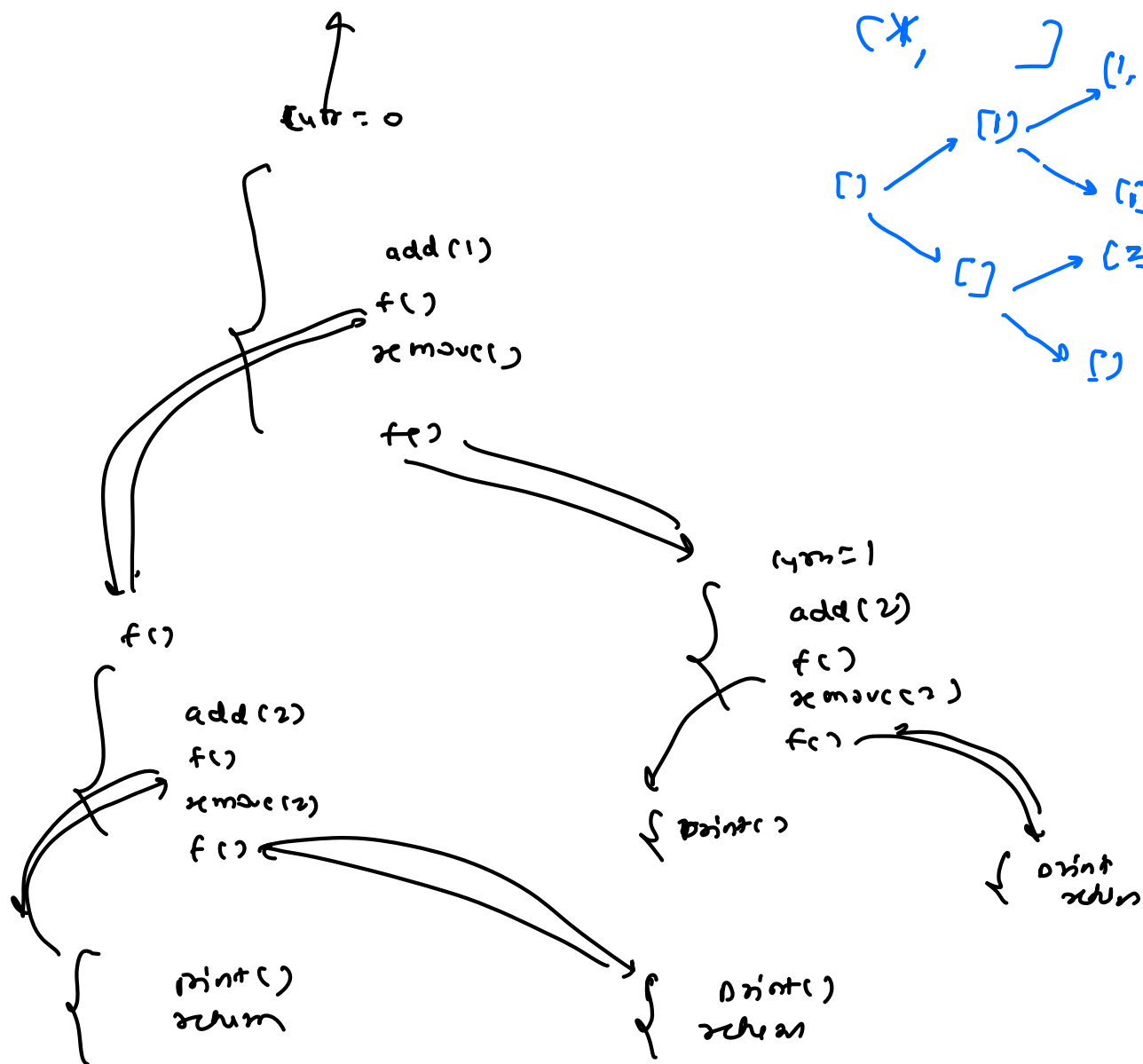
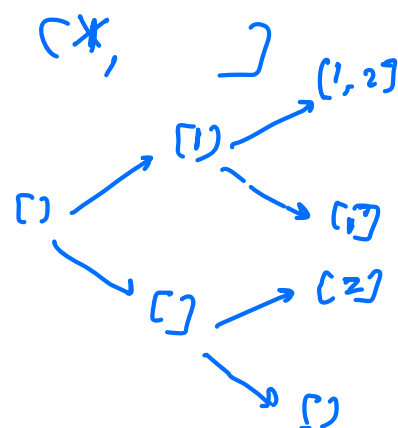
```
// option 2 Dont take
```

```
    solve ( A, curr+1, currsubset)
```

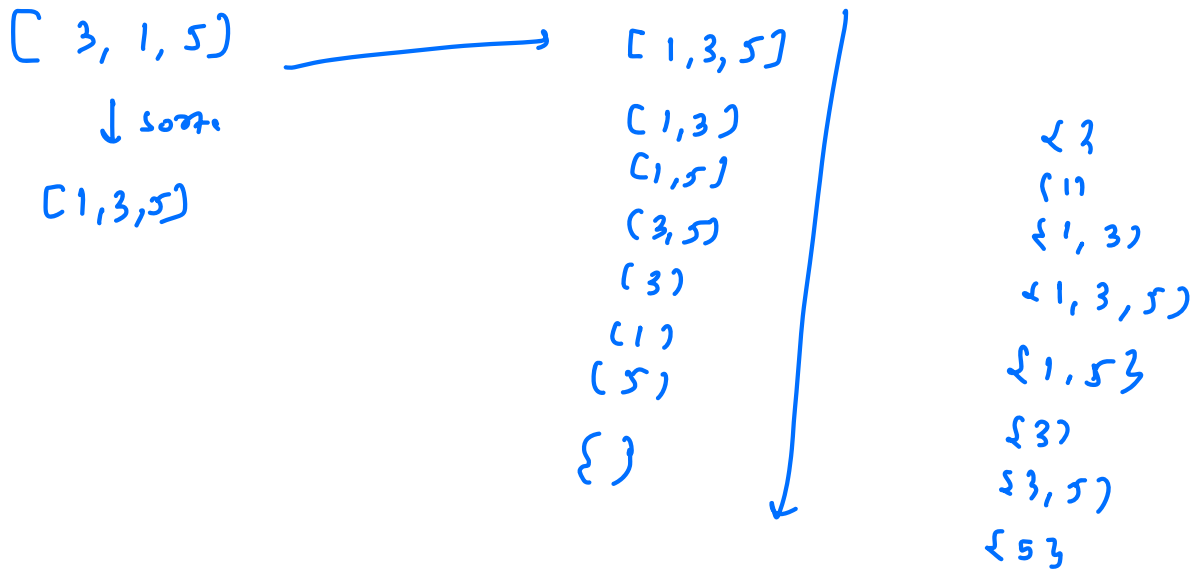


$T(n) = O(2^n)$
 $S(n) = O(n)$

~~$[x, x]$~~



$(1, 2)$
 (1)
 (2)
 (3)



Q.



No. of permutations

$$N, (N-1), (N-2), \dots = N!$$

$${}^N P_N = \frac{N!}{(N-N)!} = N!$$

distinct element

Q. Given String. Return all permutation of string

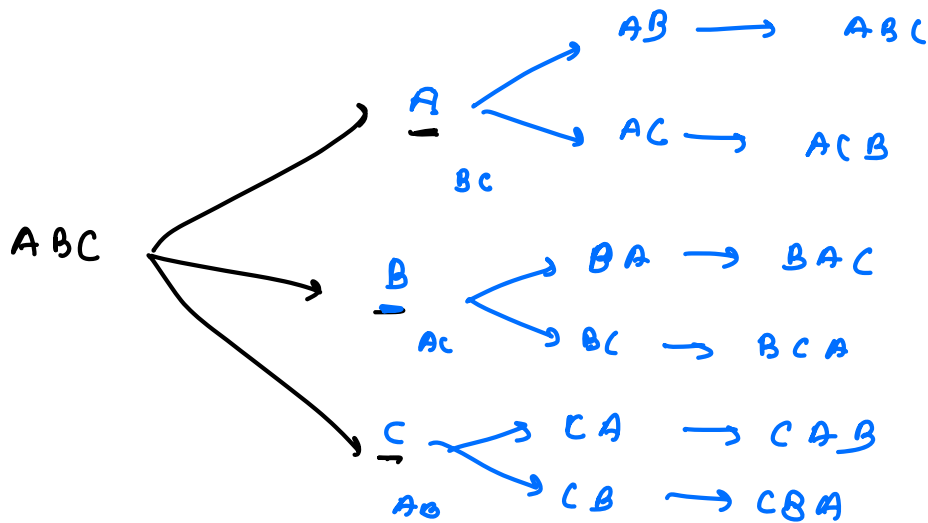
Ex: ABC

ABC
ACB
BAC
BCA
CAB
CBA

3!

String = AB

AB
BA 2!



```
void solve ( A[], int curr, ArrayList ans, ArrayList visited[N])
```

```
    if ( curr == N )
    {
        print (ans)
        return
    }
```

```
    for ( i = 0; i < N; i++ )
```

```
    {
        if ( visited[i] == false )
```

```
        {
            ans[curr] = A[i]
```

```
            visited[i] = true
```

```
            solve ( A, curr+1, ans, visited )
```

```
            visited[i] = false
        }
```

[a, b]

f(0, ans, vis)

ans [~~a~~, ~~b~~]

vis [X, X]
~~X~~ ~~X~~
~~X~~ ~~X~~
~~f~~ ~~f~~
~~f~~

for (j = 0; j < 2; j++)

if (visited[0] == false)

f ()

f(1, ans, vis)

f ()

$f \subset 2$

$[a, b]$

$[b, a]$

$TC: O(N!)$

$SC: O(N)$

x

x

~~$f(1)$~~
 ~~$f(2)$~~
 ~~$f(3)$~~
 ~~$f(4)$~~